

DevelopersHub Corporation

Task 1 Theory – Exploring and Visualizing a Dataset (Iris Dataset)

1. What is a Dataset?

- A **dataset** is a collection of data, usually organized in rows and columns (like a table).
 - Each **row** is a data point (example), and each **column** is a feature (attribute).
 - Example: In the Iris dataset, each row is a flower, and columns include `sepal_length`, `sepal_width`, `petal_length`, `petal_width`, and `species`.
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2. What is EDA (Exploratory Data Analysis)?

- **EDA** is the process of **understanding your data** before doing any analysis or modeling.
 - Goals of EDA:
 1. Check **data structure** (rows, columns, missing values, types)
 2. Get **summary statistics** (mean, min, max, standard deviation)
 3. Find **patterns, relationships, and outliers**
 4. Decide which features are important for analysis
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3. Python Libraries for Data Analysis

a) pandas

- Used for **data manipulation and analysis**
- Helps you **load, inspect, and process datasets**
- Main features used in EDA:
 - `.read_csv()` → load CSV files

- `.head()` → see first few rows
 - `.info()` → data types, missing values
 - `.describe()` → summary statistics
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b) matplotlib

- A **basic plotting library** in Python
 - Used to create **line charts, scatter plots, histograms, bar charts, and more**
 - Provides **full control** over plot style and formatting
 - Example: `plt.plot()`, `plt.scatter()`, `plt.hist()`
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c) seaborn

- Built on top of matplotlib
 - Designed for **statistical data visualization**
 - Makes plots **prettier and easier** with fewer lines of code
 - Special features:
 - Automatically colors plots based on **categories**
 - Easy creation of scatter plots, box plots, pair plots, etc.
 - Example: `sns.scatterplot()`, `sns.boxplot()`, `sns.pairplot()`
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4. Types of Visualizations in EDA

1. Scatter Plots

- Show relationship between **two numerical features**
- Can reveal **correlation or clusters**

2. Histograms

- Show the **distribution of values** for a feature
- Help find **common values or spread**

3. Box Plots

- Show **median, quartiles, min, max, and outliers**
- Useful for spotting **extreme values**

4. Pair Plots

- Show scatter plots for **all pairs of numerical features**
- Gives a **quick overview of relationships** across the dataset

5. Why EDA is Important

- Helps you **understand the data before modeling**
 - Reveals **patterns, trends, or anomalies**
 - Reduces chances of **mistakes** in later analysis
 - Makes your analysis **more reliable and interpretable**
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Easy Comparison Table (Memorize this ❤️)

Plot	Purpose	Shows	Best for	
Scatter Plot	Relationship	Pattern between 2 variables	Correlation	
Histogram	Distribution	Frequency of values	Data spread	
Box Plot	Outliers & spread	Median & quartiles	Outlier detection	

Scatter plots → relationships
Box plots → outliers

Histograms → distributions
Pair plots → multi-feature insight
